PHASE 2 PROJECT REPORT

Speed–Quality Trade-off in FreeMCG Counterfactual Generation

An Ablation Study of Reverse-Diffusion Steps

Based on the research paper

Derivative-Free Diffusion Manifold-Constrained Gradient for Unified XAI

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Prepared from the supplied Phase 2 notebook, saved experiment outputs, and reproduction guide.

Abstract

This project studies the effect of reducing the number of reverse-diffusion steps in FreeMCG counterfactual generation. The experiment compares 100, 50, and 25 steps using three ImageNet validation images: a cheetah, a rhinoceros beetle, and a tabby cat. Ladybug is used as the shared target class. For each image/step setting, 100 particles are generated and evaluated using runtime, target-class flip rate, mean target probability, and mean L2 distance. Mean runtime decreases from 35.45 minutes at 100 steps to 17.72 minutes at 50 steps and 8.86 minutes at 25 steps, corresponding to approximately 50% and 75% reductions. However, mean target-class probability also decreases, from 0.01691 to 0.00256 and 0.00227. Mean flip rate is 1.33%, 0%, and 0.33%, respectively. The results show a clear runtime benefit but do not show preservation of target-class success under the tested setup. The study is a small, controlled ablation across three selected images and should not be generalized to all ImageNet classes.

Keywords: Explainable AI; FreeMCG; counterfactual explanation; diffusion models; reverse diffusion; runtime; ablation study.

1. Introduction

Explainable Artificial Intelligence (XAI) aims to make the behavior of machine-learning models more understandable. Two common explanation goals are feature attribution and counterfactual explanation. Feature attribution identifies which input regions influence a model's prediction. A counterfactual explanation changes an input so that the model produces a specified alternative prediction.

This project is based on the CVPR 2025 paper *Derivative-Free Diffusion Manifold-Constrained Gradient for Unified XAI* by Kim and co-authors. The paper proposes FreeMCG, a derivative-free diffusion-based framework for unified XAI. This project focuses on its counterfactual-generation component and investigates its runtime–quality behavior when the number of reverse-diffusion steps is reduced.

1.1 Problem statement

Reverse-diffusion-based counterfactual generation can produce higher-fidelity results, but it can be computationally slow. The practical question is whether fewer reverse-diffusion steps can reduce generation time while retaining the ability to produce counterfactual particles that the classifier assigns to the requested target class.

1.2 Aim and objectives

• Measure how reverse-diffusion step count affects counterfactual-generation runtime.

• Compare target-class success using flip rate and mean target-class probability.

• Measure image displacement using mean L2 distance and inspect generated visual panels.

• Compare 100-, 50-, and 25-step configurations while keeping the other principal settings fixed.

• Identify limitations and define appropriate future work.

2. Original Method and Research Gap

FreeMCG uses diffusion-model information and classifier feedback to construct explanations. In counterfactual generation, the aim is to find modified images that move the classifier's prediction toward a chosen target category. The original paper discusses reverse diffusion as a route to higher-fidelity counterfactuals, while noting that this route can be slower. This creates a practical tension between computational cost and explanation quality.

2.1 Research gap selected

This project investigates a narrowly defined question: how does reducing the number of reverse-diffusion steps from 100 to 50 and 25 affect runtime and target-class success in the FreeMCG counterfactual-generation workflow? This is an experimental extension of the runtime limitation discussed in the original paper. It is not a claim that diffusion acceleration has never been studied.

2.2 Research question

Can reducing reverse-diffusion steps improve the computational efficiency of FreeMCG counterfactual generation while maintaining acceptable target-class success?

3. Proposed Improvement and Supporting Literature

The proposed improvement is a step-count ablation. The baseline condition uses 100 reverse-diffusion steps. Two reduced-step conditions use 50 and 25 steps. The experiment does not replace the FreeMCG algorithm or introduce DDIM sampling; it changes the step-count setting in the existing workflow and measures the resulting trade-off.

Supporting literature: Song, Meng, and Ermon (2021), *Denoising Diffusion Implicit Models*, studies diffusion sampling procedures that can reduce sampling cost and motivates evaluating the relationship between sampling computation and output quality. DDIM is conceptual support for studying sampling efficiency; it is not the sampler implemented in this project.

3.1 Hypothesis

Reducing the number of reverse-diffusion steps is expected to reduce runtime. Quality-related metrics may deteriorate if the reduction is too aggressive. The experiment tests this expectation rather than assuming that fewer steps are always better.

4. Methodology and Implementation

4.1 Experimental design

The experiment contains three source images, each evaluated at three reverse-diffusion step counts, producing nine image/step conditions. Each condition generates 100 particles. The same target class, ladybug, is used for all three images to keep the target condition fixed while the source image varies.

Item	Setting
Source images	Cheetah; rhinoceros beetle; tabby cat (one image per class)
Target class	Ladybug for all three source images
Reverse-diffusion steps	100 (baseline), 50, 25
Particles (K)	100 per condition
Classifier	ImageNet-pretrained ResNet-50
Diffusion model	OpenAI 256×256 unconditional diffusion checkpoint used by FreeMCG
Strength / start timestep	0.4 / 400
FreeMCG step size (alpha) / proximal weight (beta)	0.2 / 0.2
Eta	0.0
Diffusion micro-batch / classifier batch	5 / 5
Seeds	1, 2, 3 for the three images; same image-specific seed reset across step settings

4.2 Data and image selection

The selected ImageNet validation files are ILSVRC2012_val_00005006.JPEG (cheetah), ILSVRC2012_val_00003191.JPEG (rhinoceros beetle), and ILSVRC2012_val_00016389.JPEG (tabby cat). The classifier's initial predicted source class matched the requested source category for each selected image. Because there is only one image per source class, the experiment is a controlled demonstration rather than a class-level benchmark.

4.3 Why ladybug is the target for all three images

The notebook explicitly fixes ladybug as the target class. This keeps the target condition constant, allowing comparison of step counts across three different source images without also changing the target category. It is narrower than the original paper's broader evaluation protocol. Conclusions here apply only to these three source-to-ladybug cases.

4.4 Implementation workflow

• Load the FreeMCG implementation, diffusion checkpoint, ImageNet validation images, and pretrained ResNet-50 classifier.

• Apply the matching ResNet-50 preprocessing and verify that the optimized batched classifier returns logits equivalent to the reference classifier; the notebook reports a passed equivalence check.

• For each source image, run counterfactual generation at 100, 50, and 25 reverse-diffusion steps.

• Generate 100 particles for every condition while keeping the other listed parameters fixed.

• Record elapsed runtime, flip rate, mean target probability, best target probability, and L2-distance statistics.

• Save per-image CSV results, aggregate mean and standard deviation, side-by-side visual panels, and four comparison graphs.

4.5 Evaluation metrics

Metric	Meaning	Interpretation
Runtime	Elapsed time to complete one image/step condition	Lower is faster
Flip rate	Percentage of 100 particles classified as ladybug	Higher indicates more target-class successes
Mean target probability	Average ResNet-50 probability assigned to ladybug across particles	Higher indicates stronger average classifier support
Mean L2 distance	Average Euclidean distance between generated particles and original input tensor	Smaller means numerically closer; not necessarily more realistic or succ

5. Experimental Results

The values below are calculated from the supplied three_class_summary.csv. Mean and standard deviation are across the three selected images, not across independent repeated runs within each class.

Steps	Runtime (min)	Flip rate	Mean target probability	Mean L2
100	35.45 ± 0.04	1.33 ± 2.31%	0.01691 ± 0.02007	96.99 ± 17.38
50	17.72 ± 0.02	0.00 ± 0.00%	0.00256 ± 0.00302	80.37 ± 4.97
25	8.86 ± 0.02	0.33 ± 0.58%	0.00227 ± 0.00288	65.26 ± 6.55

5.1 Runtime improvement

The mean runtime falls from 35.45 minutes at 100 steps to 17.72 minutes at 50 steps and 8.86 minutes at 25 steps. This corresponds to approximately 50.0% runtime reduction for 50 steps and 75.0% for 25 steps. Runtime therefore scales almost proportionally with the tested step count under this execution environment.

5.2 Counterfactual target-class success

Mean flip rate is 1.33% at 100 steps, 0.00% at 50 steps, and 0.33% at 25 steps. Mean target probability decreases from 0.01691 at 100 steps to 0.00256 at 50 steps and 0.00227 at 25 steps. These values are low across all configurations. The 100-step setting is strongest on average for the target-class metrics, but even its success rate is weak.

5.3 L2 distance

Mean L2 distance decreases from 96.99 at 100 steps to 80.37 at 50 steps and 65.26 at 25 steps. Generated particles are numerically closer to the original image on average at lower step counts. However, lower L2 distance does not establish better counterfactual quality: the generated image may simply have moved less and may fail to reach the target class.

5.4 Per-image results

Source image	Steps	Runtime (min)	Flip rate	Target prob.	Mean L2
cheetah	100	35.49	0.0%	0.00833	115.41
cheetah	50	17.74	0.0%	0.00092	74.96
cheetah	25	8.87	0.0%	0.00064	58.48
rhinoceros beetle	100	35.41	4.0%	0.03985	80.89
rhinoceros beetle	50	17.69	0.0%	0.00604	84.74
rhinoceros beetle	25	8.84	1.0%	0.00560	71.55
cat (tabby)	100	35.44	0.0%	0.00256	94.68

Source image	Steps	Runtime (min)	Flip rate	Target prob.	Mean L2
cat (tabby)	50	17.72	0.0%	0.00071	81.40
cat (tabby)	25	8.87	0.0%	0.00057	65.76

5.5 Graphical results

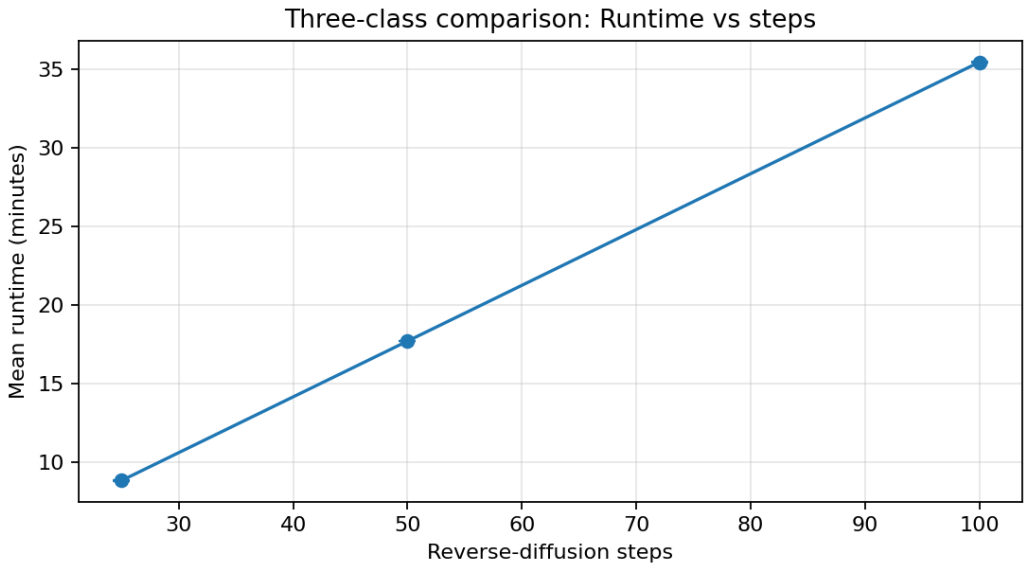


Figure 1. Runtime versus reverse-diffusion steps.

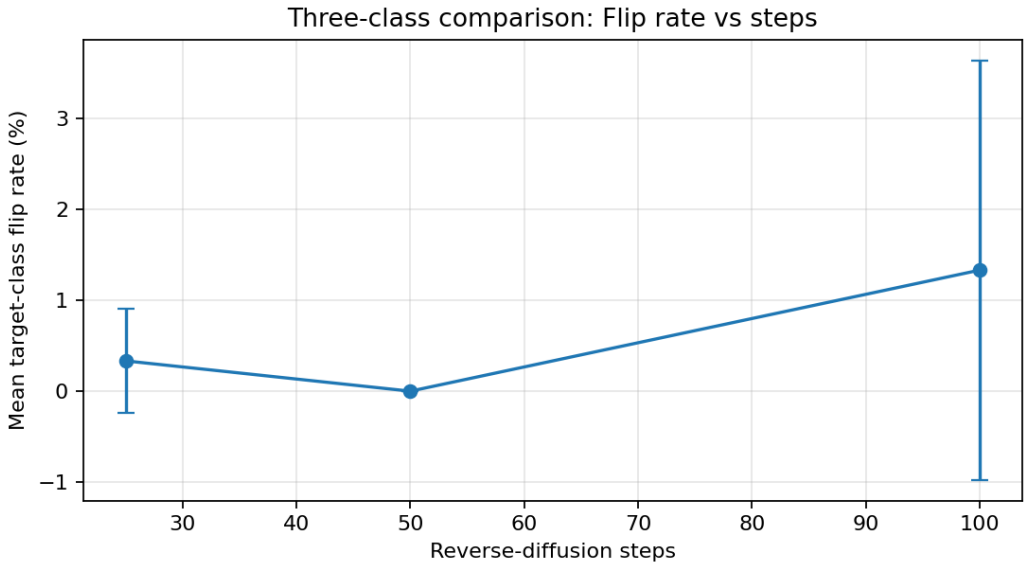


Figure 2. Target-class flip rate versus reverse-diffusion steps.

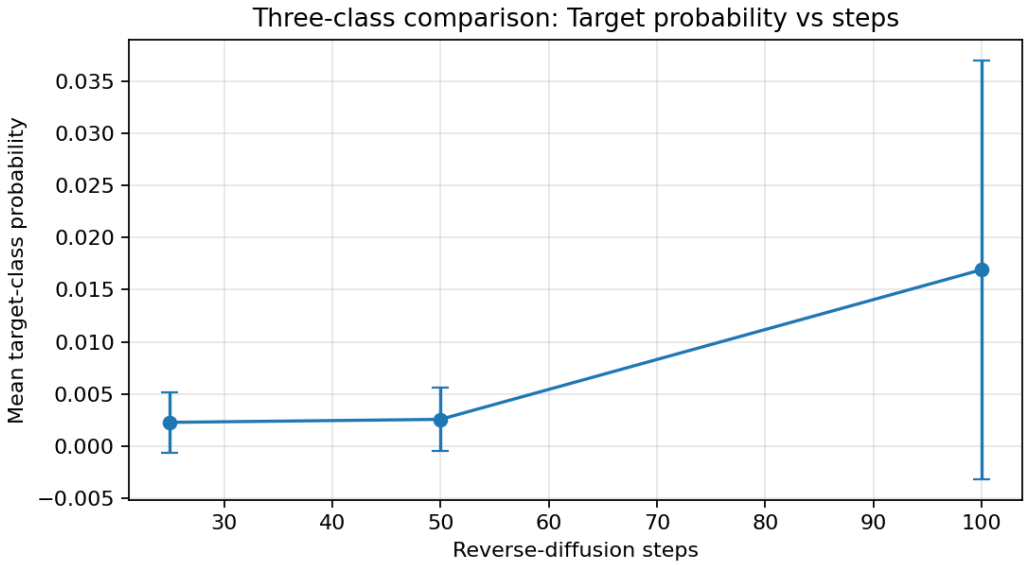


Figure 3. Mean target-class probability versus reverse-diffusion steps.

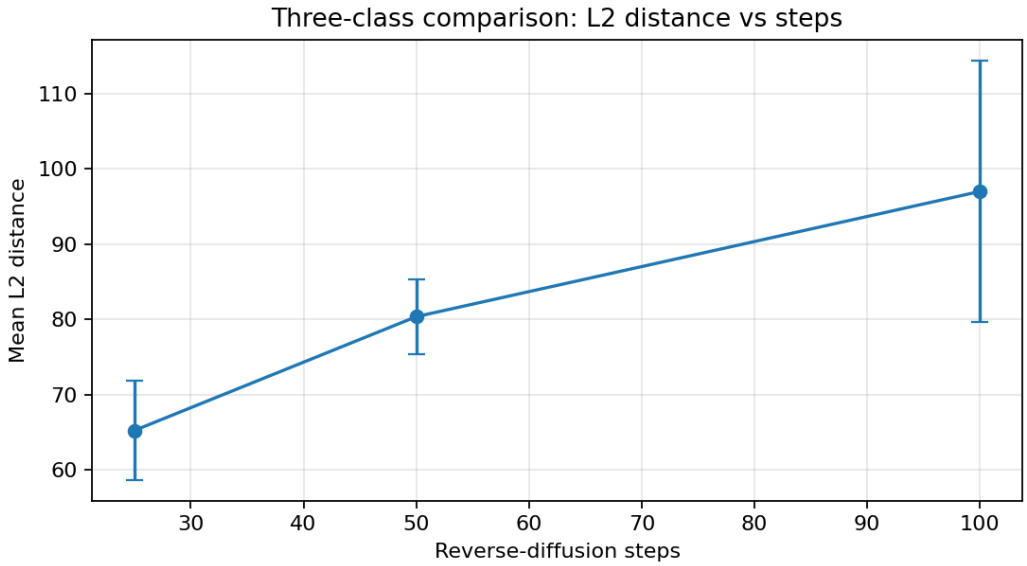


Figure 4. Mean L2 distance versus reverse-diffusion steps.

5.6 Visual comparison panels

The following panels show the original input alongside selected generated particles for each source image and step setting.

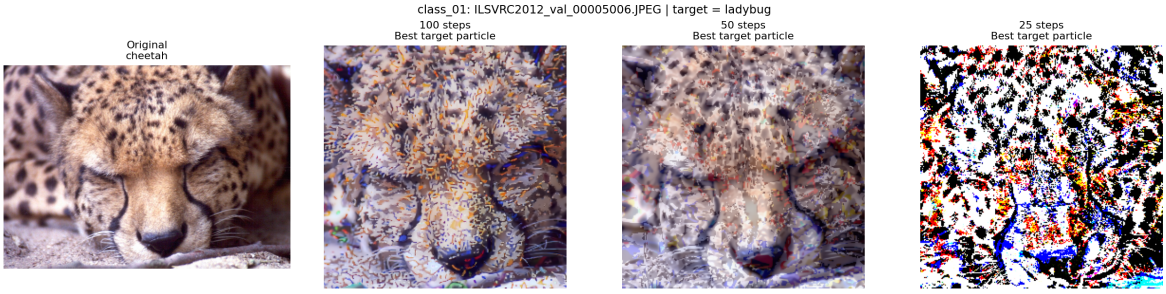


Figure 5. Cheetah source image and generated counterfactual comparison.

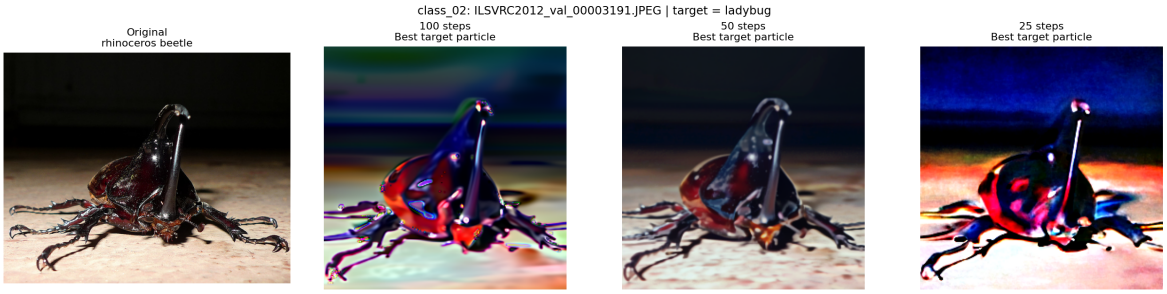


Figure 6. Rhinoceros beetle source image and generated counterfactual comparison.

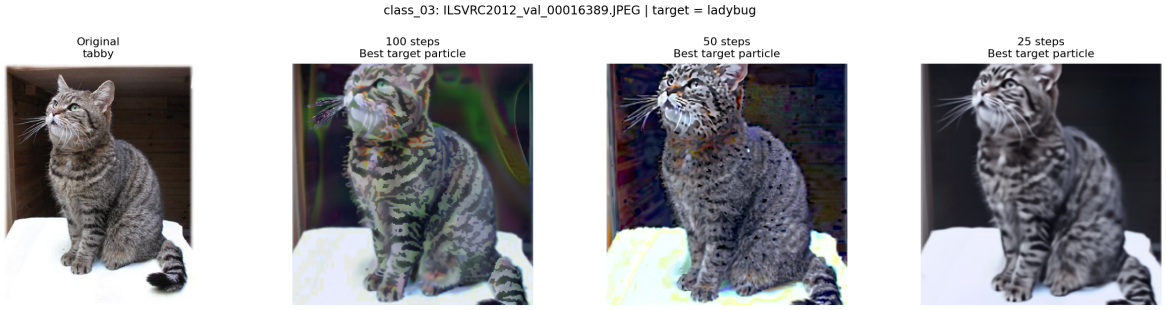


Figure 7. Tabby cat source image and generated counterfactual comparison.

6. Discussion

The experiment demonstrates a clear computational benefit from reducing reverse-diffusion steps. The 50-step condition approximately halves runtime, while the 25-step condition reduces runtime by approximately three quarters. However, the target-class metrics do not show that this speed-up preserves counterfactual success. The average flip rate is zero at 50 steps and only 0.33% at 25 steps; mean target probability is substantially lower than at 100 steps.

The results therefore do not support the claim that reduced-step generation is an overall quality-preserving improvement for the tested setup. A more accurate conclusion is that step reduction improves speed but exposes a speed–target-success trade-off. The 100-step setting performs best among the tested configurations on target probability and flip rate, although absolute target success remains low. This suggests that the target-generation setup may need further tuning or a different sampling strategy.

7. Conclusion

This project evaluated the effect of reverse-diffusion step count on FreeMCG counterfactual generation using three selected ImageNet images and a common ladybug target. Reducing the steps from 100 to 50 and 25 reduced mean runtime by approximately 50% and 75%, respectively. At the same time, mean target probability declined and flip rate remained close to zero. Thus, the tested reduction improves computational efficiency but does not preserve strong target-class success. The main contribution is a reproducible, small-scale ablation of runtime and target-success behavior under three step-count settings.

8. Limitations

• Only three images were tested, with one image per source class; results cannot be generalized to all images or classes.

• Only one target class (ladybug) was used for all source images, making the evaluation narrower than the original paper's broader protocol.

• Each image/step condition uses one fixed seed; there are no repeated independent runs for statistical inference.

• Flip rates and target probabilities are low across all settings, so the experiment does not demonstrate robust counterfactual success.

• L2 distance is a tensor-space measure and does not directly measure perceptual realism or semantic validity.

• The supplied ZIP contains saved notebook outputs and a reproduction guide, but not the complete FreeMCG repository, diffusion checkpoint, or ImageNet validation dataset; a fresh end-to-end reproduction requires those external assets.

• Runtime depends on hardware, software versions, model loading, and execution environment; it is not a universal runtime benchmark.

9. Future Work

• Evaluate more images per source class and include more source and target categories.

• Repeat each condition across multiple random seeds and report uncertainty estimates.

• Investigate intermediate step counts and alternative timestep schedules.

• Study sampling methods designed for faster diffusion, while ensuring compatibility with FreeMCG guidance.

• Add perceptual and human-evaluation measures alongside classifier success and L2 distance.

• Investigate the low target success observed even at 100 steps, including target selection and FreeMCG guidance parameters.

10. Reproducibility and Execution

The supplied reproduction guide recommends Python 3.10.x, Jupyter or Kaggle Notebooks, and an NVIDIA GPU with CUDA support. The notebook depends on PyTorch, torchvision, NumPy, pandas, Matplotlib, Pillow, tqdm, FreeMCG source modules, the OpenAI 256×256 unconditional diffusion checkpoint, ImageNet validation images, and pretrained ResNet-50 weights. Before a fresh run, configure project paths and checkpoint/data locations. The notebook's classifier equivalence check should pass before running all nine conditions. The result files include per_image_results.csv, three_class_summary.csv, three_class_summary.json, visual panels, and aggregate plots.

11. References

[1] W. J. Kim, H. Chung, J. Kim, S. Lee, B. Sim, and J. C. Ye, “Derivative-Free Diffusion Manifold-Constrained Gradient for Unified XAI,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

[2] J. Song, C. Meng, and S. Ermon, “Denoising Diffusion Implicit Models,” International Conference on Learning Representations (ICLR), 2021. arXiv:2010.02502.

[3] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.

[4] J. Deng et al., “ImageNet: A Large-Scale Hierarchical Image Database,” IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2009.

Project artifacts: Phase 2 Jupyter notebook, README reproduction guide, per-image CSV, aggregate CSV/JSON, generated comparison graphs, and visual panels supplied by the project team.